

Budget-Consistent Sea Level Component Projections with Observationally Constrained Ice Sheet Uncertainty

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Abstract

Probabilistic sea level rise projections typically sum independent draws from component-level distributions (thermal expansion, glaciers, ice sheets, terrestrial water storage), discarding the constraint that components must sum to the observed total. We demonstrate a post-processing method that enforces this budget constraint on existing FACTS v1.0 component distributions, producing revised projections that are jointly consistent with satellite-era observations. Resolving Antarctica into West Antarctic (WAIS), East Antarctic (EAIS), and Antarctic Peninsula (APIS) sub-regions, we show that the budget constraint acts as an asymmetric filter: it preferentially eliminates process-model runs that show WAIS mass gain under warming—runs inconsistent with the observed sea level budget—truncating the lower tail of the WAIS distribution and shifting the median upward. For the critical WAIS component, we separate the projection into a budget-constrained committed trajectory and a deep-uncertainty marine ice-sheet instability (MISI) component, parameterized by an activation probability $\pi(t)$. The committed component is anchored by observations; the MISI component extends the upper tail using structured expert judgment and high-end process-model output. The resulting projection distributions are provided for all SSP scenarios under multiple π assumptions, offering budget-consistent projections with transparent treatment of the WAIS deep uncertainty that dominates high-end sea level risk.

1 Introduction

2 Method

2.1 Component decomposition

2.2 Budget constraint via importance sampling

2.3 The budget constraint as a physical filter on ISMIP6

2.4 WAIS committed + MISI mixture model

3 Results

- 3.1 Budget constraint tightens component distributions
- 3.2 Asymmetric filtering of ISMIP6 WAIS distribution
- 3.3 WAIS projections with MISI uncertainty
- 3.4 Total GMSL projections
- 4 Discussion
 - 4.1 What the budget constrains and what it cannot
 - 4.2 Relationship to existing frameworks
 - 4.3 The $n = 3$ rheology bias
 - 4.4 Implications for AR7
- 5 Data and Code Availability